Institute of
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Leadership and Ideas for Tomorrow

The Impact of SME Financing Structure and Credit Composition on SME Growth - A Quarterly Macroeconomic Analysis of Pakistan's SME Sector

Research Paper

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Abstract

This study uses quarterly macroeconomic data from 2016 to 2024 to evaluate the influence of SME (Small and Medium-sized Enterprises) funding structure and composition on SME growth in Pakistan. SMEs are essential to Pakistan's economy, contributing significantly to GDP and jobs. However, they continue to confront problems, notably in obtaining formal financial services. This study fills a vacuum in empirical research by investigating how diverse types of SME credit, such as trade finance, public sector commercial banks, and private sector credit, affect SME growth at the national level.

The study uses the SARIMA and SARIMAX models to analyze the data, indicating that trade finance and public sector commercial banks considerably improve SME funding, hence increasing company activity and growth. The findings highlight the need to integrate SMEs into formal financial systems in order to boost macroeconomic growth. In contrast, the number of SME borrowers alone has little influence on growth unless enough funding is available.

The findings emphasize the need for focused policy initiatives to improve the quality and accessibility of SME finance. Expanding access to trade finance instruments, leveraging public sector banks for targeted SME lending, and creating a centralized SME credit performance monitor are among the recommendations. The study suggests that promoting long-term SME-led economic development in Pakistan requires a diverse and structured loan composition.

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Introduction

In Pakistan, SMEs constitute approximately 90% of all businesses and contribute nearly 40% to the national GDP, playing a pivotal role in employment generation and economic diversification (Dawn, 2024). Despite their significance, SMEs in Pakistan continue to struggle with persistent challenges, including limited access to credit, outdated technology, weak infrastructure, and inadequate managerial skills (Muhammad Abrar-ul-Haq et al., 2015). Globally and locally, access to finance is regarded as a key enabler of SME growth, facilitating their transition from survival to strategic development (Makena et al., 2014; Beck et al., 2008). It enhances productivity, supports innovation, reduces the cost of capital, and enables participation in both domestic and global value chains (Abor et al., 2014; OECD, 2021; World Bank, 2013). Recent developments in the SME segment raise concerns over the persistent underutilization of financial services, creating an imbalance in the financial structure and credit composition. Although SMEs reached outstanding loans up to PKR 640 billion, private sector credit accounted for as low as 4.2% (SBP, 2025) only 180,000 SME borrowers accessed credit, and out of an estimated 5.2 million operational SMEs (Economic Survey of Pakistan, 2023; SMEDA, 2022).

Purpose of Research

There is a notable absence of empirical studies that explore how the structure, volume, outreach, and purpose of SME credit influence national-level growth patterns. Few analyses have examined how different sources of credit such as private banks, public institutions, or Islamic financing impact SME performance over time (Crawford, Cui, & Kewley, 2023; OECD, 2021; Alkahtani et al., 2019). Therefore, the purpose of this study is to fill this gap by examining the macroeconomic impact of SME financing in Pakistan using national data, with a focus on how the

composition and origin of SME credit predict growth outcomes. By evaluating SME performance through this lens, the research aims to generate evidence-based insights that can guide smart policy interventions, optimize the design of targeted financial products, and help prioritize funding channels that most effectively support SME-led economic development across the country.

Literature Review

Importance of SME Financing

Financing SMEs is one of the main functions of the business finance market. Since SME growth plays a pivotal role in fostering employment opportunities and enhancing economic performance. Adequate funding for labor, capital, and inventory can help SMEs shift from survival to growth. Similarly, Ahmed and Alam (2015), support the notion where they emphasize the importance of access to finance as key to economic growth and development.

Barriers to Finance

The growing role of SMEs in driving economic development underscores the increasing importance of their access to financial services. However, small businesses face challenges that could hinder their growth and long-term sustainability (Mahmood, Yoruk, & Gilman, n.d.). SMEs frequently struggle to obtain financing through the official sector, which restricts their capacity to fully support economic expansion. This notion is also supported by Ahmad & Alam, (2015) and Goldhausen (2017) where one of the primary obstacles to SME growth is access to financing. Regardless of an economy's degree of growth, SMEs are regarded as its foundation. They are fundamental to driving economic growth and development. Therefore, as per Hallberg (2000), governments ought to support SMEs since they make up a sizable portion of all businesses in an economy, have the potential to create a significant number of jobs, and serve as "the emerging private sector in poor countries," forming a "base for private sector-led growth."

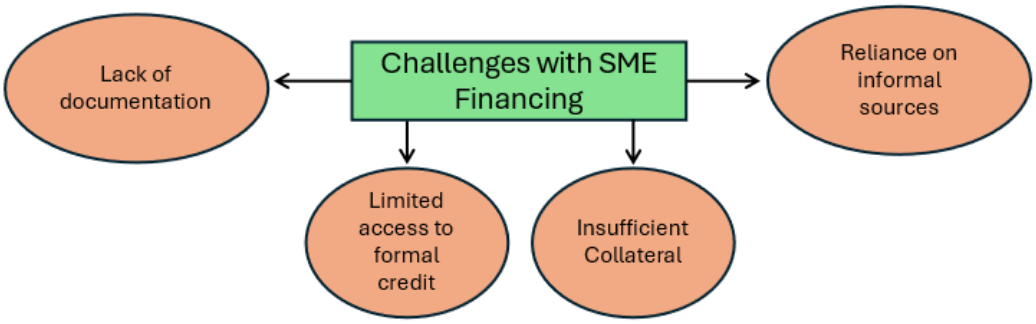
Nkurunziza (2005) also claims that the slow growth of SMEs has been ascribed to a lack of access to financial resources, while Durand's (1952) study found that SMEs can use debt financing in the form of SME loans and new credit lines to lower their

cost of capital and, as a result, raise their value. Also, The Pecking Order Theory (Myers, 1984) states that businesses favor debt financing over equity because it reduces information asymmetry and prevents ownership dilution (Badi & Ishengoma, 2021).

According to Beck et al. (2005), when examining funding barriers, challenges with banks, including bank documentation and bureaucracy, as well as the requirement to have certain relationships with banks, do limit the expansion of businesses. Significant barriers also arise from collateral requirements and specific access difficulties, such as finance for equipment leasing. Another research by Hamid and Abaidullah (2017) supports this argument where repayment issues, complicated lending procedures, and lack of personal guarantees are regarded as core reasons restricting funding. Their research also indicates that despite these obstacles, improved financing patterns and regulatory attention will probably result in the expansion of SMEs, improved management techniques, and more robust contributions to economic growth.

Companies' ability to function depends on their ability to obtain debt financing, whereas businesses and other organizations worldwide, especially SMEs, depend on timely, reasonable, and sufficient financing. Small businesses and other organizations can increase investment and launch new projects thanks to this funding (Badi & Ishengoma, 2021).

Furthermore, SMEs cannot grow to compete in international markets or even attempt to connect with larger corporations if they lack adequate funding (Harash et al., 2014). High-growth companies frequently need prompt access to outside funding to close the gap between their present investment requirements and their anticipated future income (Parsons & Titman, 2007). Due to their generally low internal capital reserves and limited capacity to obtain stock through public markets, SMEs are particularly vulnerable to this dependency (Hussain et al., 2020).



Government Interventions in SME Trade Finance

According to the OECD (2021), International commerce is linked to growth-promoting effects. It serves as a particularly significant conduit for beneficial spillovers for SMEs, including in the areas of innovation, technology, and managerial skills. International trading by SMEs can be facilitated by access to trade finance. Countries worldwide have relaxed the modalities of current assistance schemes, added new facilities, and increased support for working capital programs and export credit agencies (OECD, 2020[4]; OECD, 2020[5]). Other measures are frequently used in conjunction with these policy tools to facilitate SME access to financing in general (OECD, 2020[6]). The trade performance of SMEs can be influenced by conventional bank-intermediated trade finance instruments. These consist of letters of credit (L/C), guarantees, documentary collection, and short-term trade loans (OECD,2021).

Working Capital Management and SME Profitability

Efficient working capital management is crucial for SMEs to maintain liquidity and profitability. According to Gul et al. (2022), "a number of days accounts payable has a positive association with profitability whereas average collection period, inventory turnover and CCC have an inverse relation with performance." This highlights the Importance of how effective payables management improves profitability, which is otherwise hindered due to excessive inventory and delayed collections. According to Braimah et al. (2021) "trade receivable collection period, trade payable period, and inventory conversion period had significant effects on the performance of SMEs in Ghana," emphasizing the role of working capital components in SME performance. Ali and Ali (2014) observed that fixed capital investment has very little, if any, effect on profitability, asserting that while fixed investments are important for long-term

growth, their immediate effect on profitability may be limited.

Access to Finance and SME Growth

Evidence from research in Indonesia, MSMEs (Micro Small Medium Enterprises) that have greater access to financial services are more likely to participate in export markets and demonstrate superior business success. Similarly, SMEs in developing nations can boost their revenue, create sustainable operations, and lessen their susceptibility to outside shocks by having access to financing. According to Beck et al. (2008), such SMEs are more likely to see increases in output, employment creation, profitability, efficiency, exports, productivity, and return on assets, (Bongomin, Ntayi, Munene, & Malinga, 2017). We can also find evidence from previous studies, that Ethiopian SMEs experience improved financial inclusion due to favorable conditions in supply, demand, market opportunities, and collateral availability (Mahmood, Yoruk, & Gilman, n.d.). This was also supported by a World Bank (2013) study, where they saw how access to finance in emerging countries simplified market entry, lowered risks, encouraged innovation, and positively improved business performance (Bongomin, Ntayi, Munene, & Malinga, 2017).

Moreover, according to Nizam et al. (2020), in ASEAN countries, access to finance is positively reflected in the manufacturing sector performance, stressing the contribution of finance to increased output and enterprise development Similarly, in the retail sector, innovative financing models have proven effective; for instance, in Mexico, Verqor provides credit for agricultural inputs based on trading data, ensuring productive use of loans and minimizing default risk, with non-performing loans at 1.7% indicating the efficacy of the tailored financial products boosting SMEs.

Public Finance

Financial and non-financial incentives for SMEs have become a critical economic policy for governments around the world. This policy has included direct government grants and loans, government guarantees on loans, and a certain amount of government engagement in credit markets (Craig et al., 2011; Rubin & Ben-Aharon, 2021). In addition to providing a significant source of finance for small enterprises that are currently unable to contact venture capital firms or secure private or bank loans, these programs aim to increase the viability of SMEs in both domestic and

foreign markets (Crawford, Cui, & Kewley, 2023). Martí and Quas (2018) presented a novel model of Spanish participative loans, in which governments offer SME loans with conventional repayment terms and profit-based contributions comparable to equity. Another viewpoint by Radas and Bozic (2009) emphasized that government loan guarantees played a role in the rise of Korean venture capital (Binh et al., 2020). Furthermore, according to Hefferan and Fern (2010), Australian loan support was intended to help and nurture new businesses; similarly, SME loans from Pakistan and Romania bolstered long-term competitive performance (Alkahtani et al., 2019).

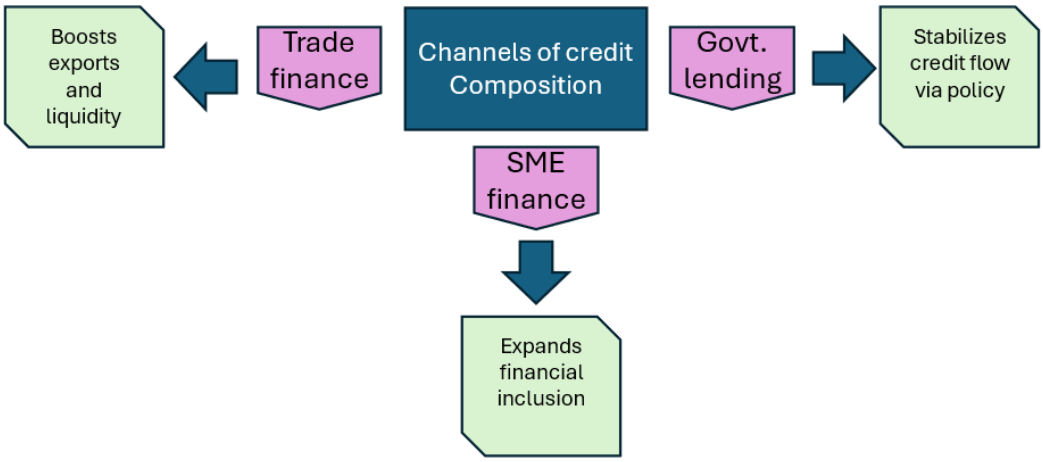
Banking (Islamic, Special & DFIs)

Over the years Islamic banking has developed into an essential channel for SME financing in Pakistan, offering Shariah-compliant products like Murabaha (cost-plus financing), Ijarah (leasing), and Musharakah (partnership-based equity financing). By upholding ethical values and equitable risk distribution, they resonate strongly with small business needs. These Islamic financial products relying on risk-sharing models have played a significant role in increasing the market share of Islamic banking assets up to 28.2% with Islamic deposits rising to PKR 8.6 trillion by the end of 2023. (Siddiqui, 2024) Moreover, empirical evidence from a study by Mushtaq et al. (2018) also supports the positive impact of Islamic finance on economic development including SME output and growth by improving financing opportunities. According to the State Bank of Pakistan (2019), DFIs are designed to serve as "catalysts for economic development by channeling funds to priority sectors including SMEs," especially in areas where commercial banks exhibit limited penetration (SBP, 2019,). The report further notes that DFIs play a key role in the "provision of long-term finance" which is critical for SME capital formation and project development. Complementing this, SMEDA (2024) identifies DFIs such as Pak Libya, Pak Oman, and Pak Brunei among the key institutions extending SME-targeted financing lines, supporting "investment-heavy ventures which commercial banks may hesitate to fund" (SMEDA, 2024,). These interventions help bridge the financing gap for small businesses, enabling fixed capital formation and scale expansion in sectors like manufacturing and services.

Specialized banks are also a pivotal financing tool for SMEs, especially in developing nations. According to Guo (2013), Zhejiang Tailong Commercial Bank

extended loans worth RMB 300,000 to more than 15,000 SMEs using relationship lending, on-site investigations third-party guarantors, and post-loan monitoring reducing credit risk and supporting collateral lacked small firms. Similarly, First City Monument Bank (FCMB) in Nigeria was also successful in raising over $50 million in IFC loan largely supporting manufacturing and agribusiness during the COVID-19 pandemic improving the business environment (IFC, 2020).

Moreover, Beck & Demirguc-Kunt (2006), believe that SMEs in developing nations can make profitable investments to grow their companies and buy the newest technology thanks to adequate financing, which guarantees their competitiveness and promotes innovation, macroeconomic resilience, and GDP growth. Thus, by encouraging greater business capitalization, job creation, and long-term income growth, Kevane & Wydick (2001) also contend that lending money to micro businesses boosts economic growth in the unorganized sector. Aghion & Bolton (1997), also support the fact that more credit leads to more firm formation, entrepreneurship, and economic growth (Bongomin, Ntayi, Munene, & Malinga, 2017).



The case for Pakistan:

Typically, in Pakistan, commercial banks prioritize investing in government securities over financing SMEs favoring higher liquidity with minimal risk. According to the SBP's Annual Report, over 60% of banks' loan portfolios will be invested in government treasury bills and bonds by the end of 2023 resulting in reduced credit availability for the SME sector This disproportionate focus on government securities restricts SMEs' access to funding, thereby hindering their

growth prospects and economic impact (State Bank of Pakistan, 2023).

In the context of Pakistan where SMEs account for about 40% of economic growth and 70% of job opportunities (SMEDA, 2018), they still face many obstacles, including high failure rates, limited financing options, antiquated technology, and a lack of managerial expertise (Muhammad Abrar-ul-Haq et al., 2015). These challenges are exacerbated by economic volatility, excessive bureaucracy, and political instability (OECD, 2008).

Additionally, limited financial literacy and advisory support further restrict SME owners from effectively navigating formal credit systems (The Express Tribune, 2024). These challenges continue to hamper the sector's ability to scale and contribute to sustainable economic growth, underscoring the need for targeted research and policy reform.

These theoretical insights support the core objective of this study: to examine how the structure and composition of SME credit—defined by its volume, outreach, purpose, and lending source—predict quarterly SME growth in Pakistan. According to the Competition Commission of Pakistan, only 6–7% of private sector financing is allocated to SMEs—far lower than in neighboring countries like Bangladesh (25%) and India (18%) (Dawn, 2023). A survey cited by the State Bank of Pakistan also revealed that 93% of SMEs find it difficult to obtain bank financing, and around 80% have never accessed credit through formal financial institutions (Business Recorder, 2023). The key barriers include rigid collateral requirements, complex loan procedures, and a lack of financial products tailored to SME needs.

Research Gap

While SMEs have always been recognized as a significant segment for an economy's growth and particularly employment generation, the majority of the research in Pakistan has been focused on micro-level analysis including firms and sectors. (Muhammad Abrar-ul-Haq 2015). Studies often highlight barriers like lack of credit access or bureaucratic delays but fall short of quantifying how different forms of SME financing impact growth at the national level (OECD, 2008; World Bank, 2013). Moreover, the informal nature of SME funding in Pakistan—largely dependent on personal savings and unregulated credit—remains underexplored in

terms of its macroeconomic implications (SBP, 2025; Khan & Qayyum, 2020). The efficacy of the impact of different credit compositions of SME credit and their impact on growth remains to be addressed. These issues highlight the necessity for in-depth research into the impact of structured financial access on the macroeconomic growth of SMEs.

Research Objective

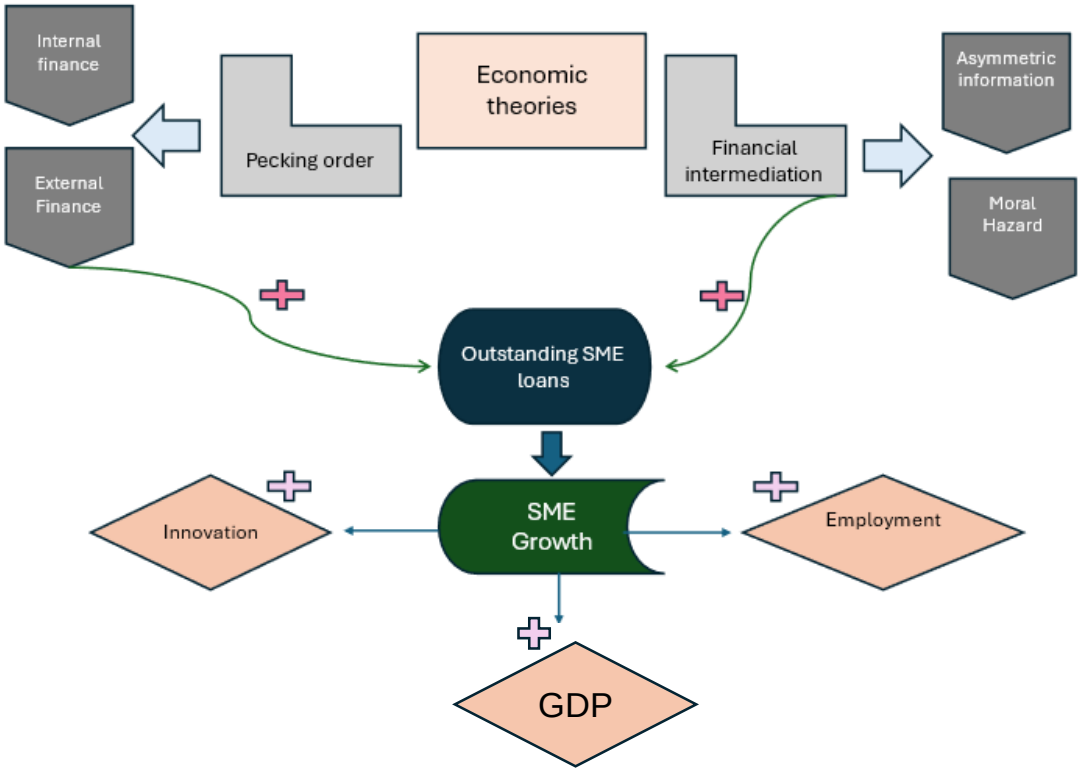
This research aims to use outstanding SME loans as a proxy for sales and business growth, analyzing quarterly changes in credit volume as an indicator of overall SME performance— based on the assumption that increased financial inflows reflect a more positive and active business environment (Beck, Demirguc-Kunt, & Maksimovic, 2008). Additionally, the study seeks to evaluate the effectiveness of various financing sources by examining their impact on the status of outstanding SME credit, thereby generating empirical evidence to inform policy interventions that promote and support diverse credit composition mechanisms (e.g., trade finance, working capital loans, fixed investment finance). A further objective is to emphasize the importance of credit availability, highlighting how increased access to financing is reflected in the expansion of outstanding SME loans (Bongomin, Ntayi, Munene, & Malinga, 2017; World Bank, 2013).

Theoretical Framework

Under the Pecking order Theory, firms always prefer internal financing amid uncertainty and information asymmetry. We can see Carpenter and Petersen (2002) argue that internal finance limits firm growth and only those business that use external funding sources are likely to expand. Thus, relying on bootstrapping can prove to be a significant growth resistor. These findings are further supported by Honjo and Harada (2006), who investigated how SME policies and firm finance structures affected SME growth. While they concluded that internal finance is a major contributor to growth, they also noted how high leverage in firms diminishes their ability to obtain external funding.

Under the financial Intermediation Theory, the financial sector including both money and capital markets play a pivotal role in the economic development. According to

Goldsmith (1969), Shaw (1973) and McKinnon (1973), the variation in economic growth among countries can be linked to the extent and effectiveness of the financial services available. Financial intermediation entails reducing the expenses related to gathering information and carrying out financial transactions. Additionally, it provides for risk pooling and insurance, and it encourages the use of credit lines to finance liquidity requirements (as cited in Shittu, 2012; Allen and Gale, 1997, 2004; Holmstrom and Tirole, 1998, Benstom and Smith. Jr, 1975) Moreover, this lowers the costs of external finance and transaction expenses generally by assisting businesses in overcoming adversity selection and moral hazard (Anigbogu et al., 2015).

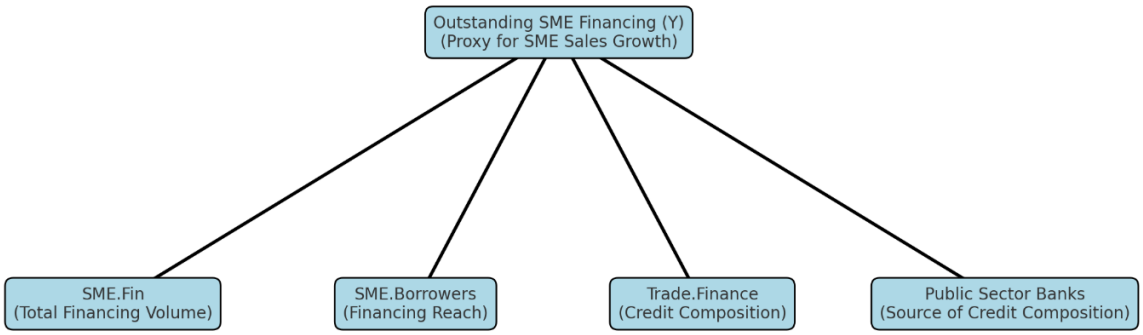


Methodology

Model Specification

Following a detailed review of previous studies and the identification of research gaps, our primary focus is to measure the impact of outstanding SME loans on SME growth by using loans as a proxy for sales growth. This research utilizes secondary data from the State Bank of Pakistan (SBP): Quarterly SME Finance Review, 2016 (Q1) – 2024 (Q4). The State Bank of Pakistan is one of the most authentic and

reliable sources of macro-level data. Our analysis will concentrate on the change in outstanding SME loans as the dependent variable, with the sources of credit composition and financing serving as our independent variables over quarterly intervals using a time series model. By using outstanding SME finance as the dependent variable (Y), we effectively capture the depth and continuity of financial access, making it a strong macro-level proxy for assessing the influence of credit on SME growth and broader economic performance in Pakistan. Several studies have discussed how financing impacts output, productivity, innovation, and export participation, all driven by sustained access to credit rather than merely credit approval (Beck & Demirguc-Kunt, 2006; World Bank, 2013).



The model of this study is stated as follows:

$$\textbf{Outstanding_SME_Financing} = \beta 0 + \beta 1 \cdot \textbf{Trade_Finance} + \beta 2 \cdot \textbf{SME_Borrowers} + \beta 3 \cdot \textbf{Public_Sector_Commercial_Banks} + \beta 4 \cdot \textbf{SME_Fin} + \varepsilon$$

Where:

- **Outstanding_SME_Financing** represents the total outstanding SME financing at time t, i.e., the total amount of SME loans disbursed that remain unpaid at the reporting date,
- **Trade_Finance** includes loans provided to SMEs to support import/export or domestic trade activities,
- **SME_Borrowers** captures the total number of individual SMEs that received financing,
- **Public_Sector_Commercial_Banks** reflects SME lending by public sector

commercial banks,

- **SME_Fin** refers to the volume of SME lending provided by privately owned domestic banks, and
- **ε** is the error term.

For sarimax model equation general form is:

$$\mathbf{Y_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \beta_3 X_{3,t} + \beta_4 X_{4,t} + \varphi_1 Y_{t-1} + \ldots + \varphi_p Y_{t-p} + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-}_q + \varepsilon_t}$$

Where:

- $\mathbf{Y_t}$ is the dependent variable at time *t*
- $\mathbf{X_{1,t}}$ **to** $\mathbf{X_{4,t}}$ are exogenous regressors at time *t*
- $\boldsymbol{\beta_0}$ **to** $\boldsymbol{\beta_4}$ are coefficients of exogenous variables
- $\boldsymbol{\varphi_1}$ **to** $\boldsymbol{\varphi_p}$ are autoregressive (AR) terms
- $\boldsymbol{\theta_1}$ **to θ_q** are moving average (MA) terms
- $\boldsymbol{\varepsilon_t}$ is the error term (white noise) at time *t*

Presentation of empirical findings

These are the variables listed below for this research:

Variable Name	Description
Date	**The quarter and year of the observation (e.g., 2016 Q1).**
Outstanding SME Financing (Y)	**The total amount of credit (in billions of PKR) outstanding to SMEs at the end of each quarter.**
SME Fin as %age of Domestic Pvt Sector Financing	**The percentage share of SME financing relative to the total private sector financing by banks. Indicates the SME sector's access to financial resources.**
SME NPLs Ratio	**Non-Performing Loan (NPL) ratio for SMEs – the**

Variable Name	Description
	proportion of SME loans classified as non-performing.
No. of SME Borrowers	**The number of unique SMEs that have taken loans in that period.**
Fixed Investment	**Loans disbursed to SMEs specifically for long-term fixed asset investment (e.g., machinery, buildings).**
Working Capital	**Loans used by SMEs to finance day-to-day operational expenses.**
Trade Finance	**Loans are provided to SMEs to support import/export and other trade-related activities.**
Trading SMEs	**Number of SMEs operating in the trading sector that have received financing.**
Manufacturing SMEs	**Number of SMEs in the manufacturing sector that received financing.**
Services SMEs	**Number of SMEs in the services sector that received financing.**
Domestic Private Banks	**Total SME financing extended by domestic privately-owned banks.**
Public Sector Commercial Banks	**SME financing is extended by publicly owned commercial banks.**

Variable Name	Description
Islamic Banks	**SME financing is disbursed by Islamic banks following Shariah principles.**
Specialized Banks & others	**Financing is provided by specialized financial institutions and other non-traditional banks.**
DFIs	**Development Finance Institutions' contributions to SME financing. These are institutions established to provide targeted financial support for development sectors, including SMEs.**

Firstly an issue of multicollinearity between explanatory variables will exist; therefore, we have dropped several variables and are dealing with those where the range is between -0.55 to 0.55 as shown in the heat map below. This would help narrow down our dependent variables in case of strong intra-correlation.

	SME Fin	*SME Borrowers*	*Trade Finance*	*Public Sector Commercial Banks*
SME Fin	1			
SME Borrowers	0.28486	1		
Trade Finance	0.55295	0.29436	1	
Public Sector Commercial Banks	-0.42682	-0.05743	-0.53573	1

Fig 1. Heat Map

The variable SME Fin is transformed from a percentage to a number by multiplying it by 100 and SME borrowers are transformed by dividing each by 10,000. This was done so that each column lies on the same scale.

For the scope of this research, we intend to use SARIMA (Seasonal Autoregressive Integrated Moving Average) and SARIMAX (Seasonal Autoregressive Integrated Moving Average with Exogenous Regressors) models. These models are widely applied in economic and SME-related time series research due to their effectiveness in capturing seasonal patterns and incorporating external variables to enhance forecasting accuracy (Zhang et al., 2018; Hyndman & Athanasopoulos, 2018). So we started off with our time series being white noise or random walk. Figure 2 below of the Box-Ljung test shows the raw data is not white noise and contains significant autocorrelation.

```
        Box-Ljung test

data:  sme_ts
X-squared = 18.677, df = 4, p-value = 0.0009096
```

Fig 2. Box-Ljung test (For White Noise)

For a random walk, these three assumptions were satisfied:

1. **No specific mean**
2. **No specific variance**
3. **Strong dependence over time**

Plot 1: The series shows no clear mean reversion – instead it wanders upwards and downwards, with no fixed trend or return to a baseline. This drifting pattern is a classic signature of a random walk.

Plot 2: The rolling mean (green line) does not stay flat — it drifts up and down across the series. The rolling variance (seen via the ±1 SD bands in Plot 2) is also non-constant, especially increasing around 2020–2022. In a stationary process, the mean and variance should remain roughly constant over time. Since both drift here, it suggests non-stationarity, often caused by a random walk.

Plot 3: High autocorrelation at lower lags, especially lag 1 (ACF ≈ 0.5+), with slow decay. This somewhat persistent memory supports random walk behavior.Seasonality and trend exist according to the Figure 3 observations,

therefore will apply a random walk with a drift.

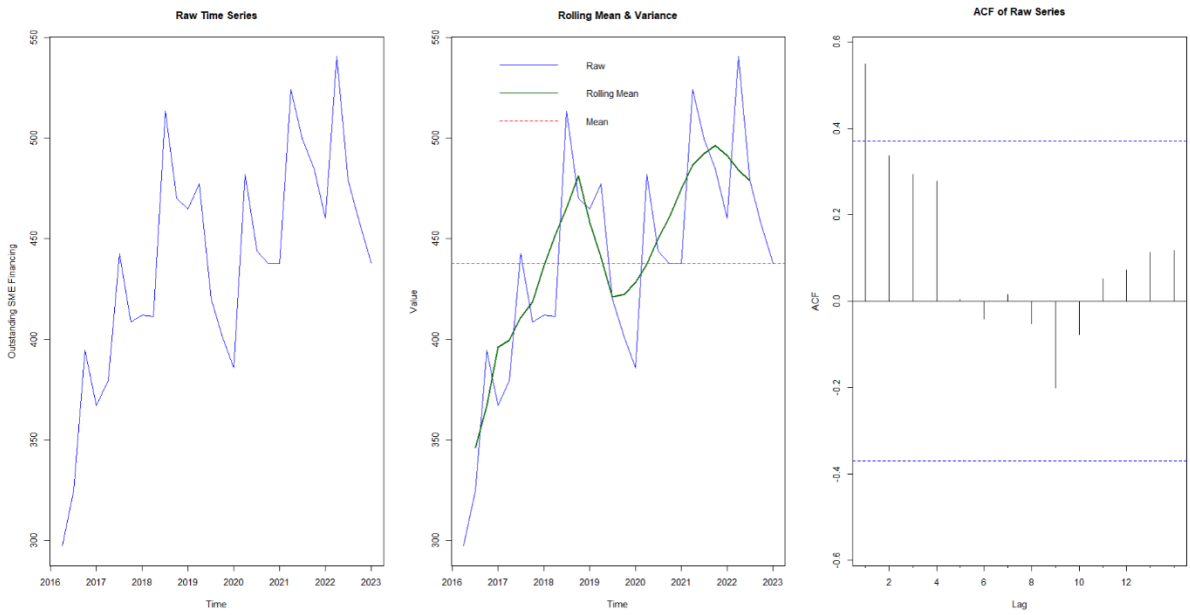


Fig 3. Random Walk

Presentation of Results

SARIMA (Seasonal ARIMA):

To capture seasonality, we cannot use the ARIMA model, we have to capture it using the SARIMA model and allow it for a drift to also cater for trend.

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	4.376076	40.43035	29.25586	0.8196575	6.653401	0.5485045	-0.1850162	NA
Test set	12.060802	31.13249	22.32914	2.0792351	4.392185	0.4186386	0.1216343	0.2901272

Fig 4. SARIMA Model

The model is statistically sound, generalizes well**,** and performs better on key metrics — especially Theil's U, MAPE, and MASE.

Test performance is stronger and more consistent, especially with:

- **MAPE under 5%**
- **MASE < 0.5**
- **Theil's U = 0.29**

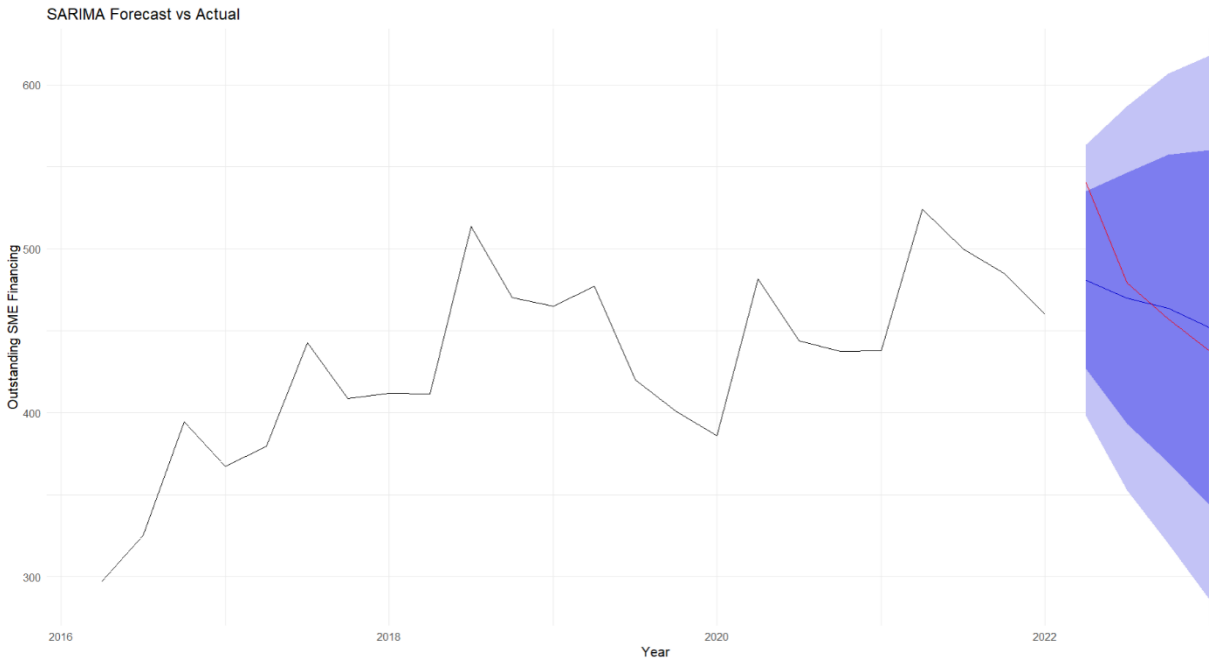


Fig 5. Check on the SARIMA Model

Model Fit: The historical pattern (black line) is well-modeled, showing seasonal fluctuations and trend shifts. The forecast (blue line) smoothly continues this trajectory — evidence that SARIMA captured both trend and seasonality.

Forecast Accuracy: The red line (actual test data) falls mostly within the blue-shaded prediction intervals — this shows the model performed accurately in forecasting. Small deviations are expected and acceptable, especially given the tight confidence intervals.

Prediction Uncertainty: The widening cone of blue shading in future periods reflects increasing uncertainty as the forecast horizon extends. This is typical in time series models and doesn't indicate model weakness — just natural forecasting limitation.

SARIMAX Model:

To Cater to exogeneity, we will use the SARIMAX Model to interpret our effect that to what extent does the structure of SME financing and credit and composition predicts quarterly SME growth in Pakistan.

```
      Coefficient   StdError          z           p
ar1     0.9890779 0.01694246 58.3786661 0.0000000000
xreg1   1.9543396 0.76478341  2.5554158 0.0106061018
xreg2  -2.2753241 3.95169471 -0.5757844 0.5647609367
xreg3   1.9521680 0.18218630 10.7152299 0.0000000000
xreg4  19.3918050 5.16505287  3.7544253 0.0001737397
```

Fig 6. SARIMAX Model

Evaluation of Results

Variable	Coefficient	p-value
Trade.Finance	1.954	0.0106
SME.Borrowers	-2.275	0.5648
Public.Sector.Commercial.Banks	1.952	0.0000
SME.Fin	19.391	0.00017

The study's methodological framework, which uses the SARIMAX model to analyze the influence of credit composition on SME growth in Pakistan, is consistent with the findings, which provide good empirical support for the hypotheses. Using Outstanding SME Financing as the dependent variable and key exogenous regressors (Trade Finance, SME Borrowers, Public Sector Commercial Banks, and SME Financing as a share of Private Sector Credit), the SARIMAX model demonstrated both statistical robustness and theoretical alignment. Trade Finance **(β = 1.954, p =.0106)** and Public Sector Commercial Banks **(β = 1.952, p <.0001)** have significant positive effects on SME financing, highlighting their critical roles in enhancing liquidity and access (OECD, 2021), consistent with the Financial Intermediation Theory (Goldsmith, 1969; McKinnon, 1973; Shaw, 1973). The variable SME Financing as a percentage of private credit has the highest impact **(β = 19.391, p =.00017)**, emphasizing the importance of integrating SMEs into formal financial structures for macroeconomic growth (Beck & Demirguc-Kunt, 2006). In contrast, SME

Borrowers **(β = -2.275, p =.5648)** are statistically insignificant, suggesting that the number of borrowers alone does not equate to improved performance unless supported by sufficient financing. This insight is consistent with the Pecking Order Theory (Myers, 1984) and empirical work by Carpenter and Petersen. These results are supported by preliminary diagnostics in the methodology, such as the Box-Ljung test and random walk validation, which validated the suitability of using SARIMAX. Overall, the findings highlight the policy importance of a diverse and structured loan composition for driving long-term SME-led economic development in Pakistan.

Conclusion

This study investigates the impact of various SME credit components on SME development in Pakistan, using SARIMA and SARIMAX models to analyze quarterly data from 2016 to 2024, with outstanding SME loans serving as a growth proxy. Our findings show that trade finance and public sector commercial banks play an important role in increasing SME financing, emphasizing their relevance in promoting company activity and expansion. Most notably, SME financing as a share of private sector credit had the greatest positive impact, emphasizing the necessity of incorporating SMEs into formal financial systems. These findings are consistent with previous research, which has emphasized the importance of accessible, structured, and institutionally backed loans in fostering SME development and closing funding gaps. In contrast, the number of SME borrowers may have a lesser or insignificant link with financing results, eliminating the negative relationship with outstanding SME loans. These findings highlight the need for a more targeted strategy for SME financing, in which banks prioritize successful lending channels such as trade finance and use public-sector engagement to boost financial inclusion. To drive long-term economic growth, policymakers should move beyond simple borrower indicators and focus on improving the quality and accessibility of SME lending.

Policy Recommendations

1. Expand Access to Trade Finance Instruments:

- Encourage banks to diversify and simplify trade finance offerings for SMEs through tailored instruments such as letters of credit, invoice discounting, and export guarantees.

2. Utilize Public Sector Banks for Targeted SME Lending:

    - Scale up SME-specific lending programs in public sector commercial banks with streamlined procedures and risk-sharing mechanisms in collaboration with DFIs.

3. Develop a Centralized SME Credit Performance Dashboard:

    - Launch a national dashboard to monitor loan type, credit source, and performance indicators, enabling more informed policy adjustments.

4. Institutionalize Targeted SME Lending Programs with Incentive

    - Develop tailored SME financing schemes through public and private banks, supported by incentives such as credit guarantees and regulatory benefits to encourage increased SME lending.

Limitations

The study uses aggregate SME loan data per quarter and does not distinguish between loans to new vs. existing SMEs. Moreover, firm-level variables like sector, size, or repayment behavior (defaults) are not included. It is also to be noted that the number of variables (sources of credit) are limited based on the data and may not include any other forms of SME credit i.e informal credit, self-help groups, microfinance, etc. Time lags associated with the successful completion of financing are not incorporated, which may affect real-time results. Lastly, Outstanding SME loans is used as a proxy for sales growth and may not perfectly equate SME Growth, whereas using actual sales data and profitability ratios associated with financing during that interval should present more meaningful outcomes to gauge growth in SMEs.

Potential Areas for More Research

Explore the direct link between SME finance and performance indicators such as debt ratios, sales growth, and profitability using firm-level data for a more comprehensive and accurate analysis of SME performance. Additionally, conduct primary, macro-level studies to account for undocumented cash flows and informal financial activity in Pakistan's SME landscape. Furthermore, investigate the role of government incentives like tax relief and subsidies on SME performance, including behavioral response and long-term sustainability.

Glossary

Abbreviation/Term	Full-Form/Explanation
SMEs	Small and Medium-sized Enterprises
OECD	Organisation for Economic Co-operation and Development
SBP	State Bank of Pakistan
NPL	Non-Performing Loan
DFIs	Development Finance Institutions
Pecking Order Theory	A theory that suggests firms prefer internal financing and will only seek external financing as a last resort due to information asymmetry and uncertainty
Financial Intermediation Theory	A theory that emphasizes the role of financial intermediaries in reducing transaction costs, pooling risks, and facilitating liquidity
Bootstrapping	Using personal savings and resources to fund a business rather than seeking external financing
Macroeconomic	Relating to the branch of economics concerned with large-scale or general economic factors, such as interest rates and national productivity

Micro-level analysis	**Analysis focused on individual firms or sectors rather than the economy as a whole**
Bureaucratic delays	**Delays caused by the complex procedures and red tape in government or organizational processes**
Unregulated credit	**Credit that is not controlled or supervised by regulatory authorities**
Structured financial access	**Organized and regulated means of obtaining financial resources**
SARIMA	**Seasonal Autoregressive Integrated Moving Average, a statistical model used for analyzing and forecasting time series data**
SARIMAX	**Seasonal Autoregressive Integrated Moving Average with Exogenous Regressors, an extension of SARIMA that includes external variables to improve forecasting accuracy**
Random Walk	**A statistical phenomenon where a variable follows a path that is seemingly random and unpredictable, often used to describe stock prices or other financial data**
Autocorrelation	**The correlation of a signal with a delayed copy of itself, often used to identify patterns in time series data**

Rolling Mean	**A statistical measure that calculates the average of a data series over a specified number of periods, used to smooth out short-term fluctuations and highlight longer-term trends**
Rolling Variance	**A measure of the variability of a data series over a specified number of periods, used to assess the stability of the series over time**
Theil's U	**A statistic used to measure the accuracy of a forecast model, with lower values indicating better performance**
MAPE	**Mean Absolute Percentage Error, a measure of prediction accuracy in a forecasting model expressed as a percentage**
MASE	**Mean Absolute Scaled Error, a measure of the accuracy of a forecast model, scaled to the error of a naive forecast**
CCC	**Cash Conversion Cycle, a measure of the time it takes for a company to convert its investments in inventory and other resources into cash flows from sales**

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